Foundations for nighttime lights data analysis

8 September 2026

Summary

A remarkable development in alternative data is the use of satellites that measure nighttime light radiance. From the early 1990s, Nightlights (NL) has been used to measure economic activity. NL correlates well with standard measures of growth and development (Doll et al. 2006), which allows measures of economic growth where existing measures are either unreliable or lack the continuity and/or resolution of constant satellite observation.

Raw monthly NL files are not analysis-ready. They contain missing observations, negative values, outliers, background noise, and cloud-related attenuation (Uprety et al. 2019; Jiang et al. 2017; Ma et al. 2014). As a result, every research team must individually implement preprocessing, often with project-specific scripts. This creates entry barriers and weakens comparability and reproducibility across studies.

Statement of need

`NighttimeLights.jl` is a Julia package for cleaning and bias-correcting monthly nighttime lights from the Visible Infrared Imaging Radiometer Suite (VIIRS). It provides a transparent and reproducible workflow that implements established methods from the applied literature, including cloud-related attenuation correction (Patnaik et al. 2021; Pearson 2002; Gupta et al. 2017).

The package addresses a practical gap: many papers using NL data rely on individual preprocessing pipelines, making it difficult to reproduce findings or compare outcomes across studies. By consolidating key cleaning steps into one standard package, `NighttimeLights.jl` reduces duplicated effort and standardizes core data preparation.

State of the field

There are two dominant nighttime-lights products used in applied work: annual DMSP-OLS composites (1992-2013) and monthly VIIRS composites (2012-

present). VIIRS has finer temporal and spatial resolution and is now the primary data source for contemporary analysis.

Most empirical NL workflows use combinations of missing-value recoding, outlier removal, and cloud-aware filtering, but implementation details vary across projects. This heterogeneity limits comparability. `NighttimeLights.jl` standardizes these operations while retaining modular control for users.

Several software tools already serve the nighttime-lights community, but they address a different part of the workflow. `blackmarbler` (Marty and Stefanini Vicente 2024) and its Python counterpart `BlackMarblePy` (Stefanini Vicente and Marty 2024) download NASA's Black Marble product and compute zonal statistics for administrative regions; `Rnightlights` (Njuguna 2019) does the same for the NOAA DMSP-OLS and VIIRS composites; and the VIIRS collections in Google Earth Engine (Gorelick et al. 2017) provide server-side access and aggregation. These tools focus on acquisition and spatial aggregation, and the Black Marble product applies sensor, terrain, and atmospheric corrections upstream at the pixel level. None of them implement a configurable statistical cleaning pipeline for the raw NOAA monthly composites, and none correct the cloud-cover attenuation bias that dominates monthly series in monsoonal regions. Contributing these routines to the existing packages would mean maintaining the same methods across three language ecosystems; providing them as a standalone Julia package, with the cleaning pipeline itself as the unit of reuse, keeps the methods in one place and keeps a cleaned panel reproducible from a single package version.

Software design

`NighttimeLights.jl` is implemented in Julia and built on `Rasters.jl` for geospatial data handling. Data is represented as raster data cubes with dimensions (x, y, t), and cleaning is organized as composable functions. Core functions include:

- Missing-data handling: `na_recode`, `na_interp_linear`
- Negative-value handling: `replace_negative`
- Outlier treatment: `outlier_hampel`, `outlier_variance`
- Background-noise filtering: `bgnoise_PSTT2021`
- Cloud-bias correction: `bias_PSTT2021`

These are wrapped into high-level pipelines such as `PSTT2021()`, `PSTT2021_conventional()`, and `clean_complete()`. This design supports both method-level customization and standardized cleaning pipelines.

Several design choices shaped the package. Primarily, cleaning is exposed as a set of small composable functions rather than a single opaque routine. This enlarges the API surface and requires the user to respect an ordering between steps, but it allows for customization and lets each stage be run, inspected,

and switched out independently, which is what makes the package usable for methodological work while still allowing reproducibility. Secondly, the package is written in Julia because the core operations are per-pixel time-series computations over large pixel counts. Hampel filters, smoothing-spline fits, linear interpolation are awkward to express as fast vectorised array code and would otherwise require C extensions, if not for Julia's loop performance. Finally, existing best-practices cleaning and cloud-bias-corrected cleaning are kept as separately named and dated pipelines (`PSTT2021_conventional` and `PSTT2021`) so that results remain reproducable as well as directly comparable to the large body of prior work that does not correct for cloud cover.

Research impact statement

The package has been used as infrastructure for empirical work where cloud cover and data quality strongly affect inference from nighttime lights. Its main impact is methodological consistency: researchers can move from raw files to analysis-ready panels using explicit, shared preprocessing steps.

Because each stage is exposed as a public function, `NighttimeLights.jl` also serves as a platform for methodological development. Users can inspect intermediate outputs, compare alternative cleaning choices, and contribute new preprocessing methods while preserving reproducible workflows.

The cloud-bias correction implemented here was developed and validated in Patnaik et al. (2021), and `NighttimeLights.jl` is the reference implementation of that method. The package has been used to build analysis-ready VIIRS panels for applied economic work, including a study of wartime economic activity in Ukraine (Hande et al. 2025) and an analysis of regional economic patterns across Java, Indonesia (Aviliani et al. 2026). It also underpins the public India Built and Lit dataset, which distributes cleaned nighttime-lights panels for India (xKDR Forum 2026). Reproducibility is supported concretely: a Mumbai example cube ships with the package and is loaded with `load_example()`, every pipeline stage is a documented public function, the package is registered in the Julia General registry, and the results quoted in this paper were produced with the pinned versions listed under Computational details.

Computational details

Results referenced in this manuscript were generated with Julia 1.12.6 and `NighttimeLights.jl` 0.9.1. Version 0.10.0, described here, changes fewer than 0.1% of the cleaned values in the bundled Mumbai example. Julia is available at https://julialang.org/ and the package is available at https://github.com/xKDR/NighttimeLights.jl.

Installation from the General registry:

```
] add NighttimeLights
```

Usage example

The following Julia code shows how to go from raw to cleaned data with `NighttimeLights.jl`. The example uses monthly VIIRS composites for a rectangle around Mumbai, January 2014 to January 2018, downloaded from Google Earth Engine by the replication notebook at https://github.com/xKDR/NighttimeLights.jl/blob/main/submission/SRC/replication.ipynb. Two inputs are needed. Both are `Raster` data cubes with dimensions (x, y, t), where t indexes months:

- `mumbai_radiance`: the monthly mean radiance of each pixel, in $nW\ cm^{-2}\ sr^{-1}$.
- `mumbai_n_cloudfree`: the number of cloud-free observations that went into each monthly mean.

The simplest path is a single call:

```
mumbai_cleaned = clean_complete(mumbai_radiance, mumbai_n_cloudfree, bgthreshold=4)
```

which is presently identical to:

```
mumbai_cleaned = PSTT2021(mumbai_radiance, mumbai_n_cloudfree, bgthreshold=4)
```

Both wrap the sequence of steps below. Each step is a public function, so the sequence can be run one step at a time, inspected, or modified:

```
# 1. Missing data: months with no cloud-free observations become `missing`
radiance_na = na_recode(mumbai_radiance, mumbai_n_cloudfree)

# 2. Negative values: recoded as `missing`
radiance_nonneg = replace_negative(radiance_na)

# 3. Background noise: mask out pixels that are dark in the annual image
lit_mask = bgnoise_PSTT2021(radiance_nonneg, mumbai_n_cloudfree, 4)
radiance_lit = apply_mask(radiance_nonneg, lit_mask)

# 4. Cross-sectional outliers: mask out pixels with extreme variance over time
stable_mask = outlier_variance(radiance_lit, lit_mask)
radiance_stable = apply_mask(radiance_lit, stable_mask)

# 5. Time-series outliers: Hampel filter on each pixel's time series
radiance_despiked = long_apply(outlier_hampel, radiance_stable)

# 6. Cloud bias: correct attenuation in months with few cloud-free observations
pixel_mask = lit_mask .* stable_mask
radiance_corrected = bias_PSTT2021(radiance_despiked, mumbai_n_cloudfree, pixel_mask)
```

```
# 7. Gaps: linearly interpolate the remaining `missing` values
mumbai_cleaned = long_apply(na_interp_linear, radiance_corrected)
```

Two helpers recur: `apply_mask` removes the pixels a 2D mask marks as `missing` from every month, and `long_apply` applies a function written for one pixel's time series to every pixel. The steps are:

1. **Missing data.** NOAA reports zero radiance for a pixel-month with no cloud-free observations, which is indistinguishable from a dark pixel. `na_recode` recodes these as `missing`.
2. **Negative values.** These arise from calibration in the presence of airglow (Uprety et al. 2019) and are recoded as `missing`.
3. **Background noise.** Unlit areas such as oceans, forests and deserts still show a small positive radiance. `bgnoise_PSTT2021` averages the last twelve months, weighting each by its cloud-free observations (Gupta et al. 2017), and marks pixels below the threshold (here 4 $nW\ cm^{-2}\ sr^{-1}$) as dark.
4. **Cross-sectional outliers.** Fires and gas flares produce pixels with extreme, highly variable radiance. `outlier_variance` marks the top 0.1% of lit pixels by the standard deviation of their detrended series as unstable.
5. **Time-series outliers.** `outlier_hampel` removes isolated spikes with a Hampel filter (Pearson 2002), using a 5-month window and a cutoff of 3 standard deviations.
6. **Cloud bias.** Cloudy months show attenuated radiance: the monsoon dip in the figure below. For each pixel kept by steps 3 and 4, `bias_PSTT2021` fits a smoothing spline of detrended radiance on the number of cloud-free observations, and raises cloudy months to the level the spline predicts for clear months (Patnaik et al. 2021).
7. **Remaining gaps.** `na_interp_linear` fills gaps by linear interpolation; leading and trailing gaps take the nearest observed value.

Steps 1 to 5 and 7, without the bias correction, are what most applied work already does. They are termed conventional cleaning in Patnaik et al. (2021) and are available as `PSTT2021_conventional()`.

An example

Figure 1 shows the aggregate raw and cleaned radiance of Mumbai, in $nW\ cm^{-2}\ sr^{-1}$, from January 2014 to January 2018; the notebook that produced it is at https://github.com/xKDR/NighttimeLights.jl/blob/main/submission/SRC/replication.ipynb. The monsoon dip (e.g. July) in the raw data is much reduced after cleaning.

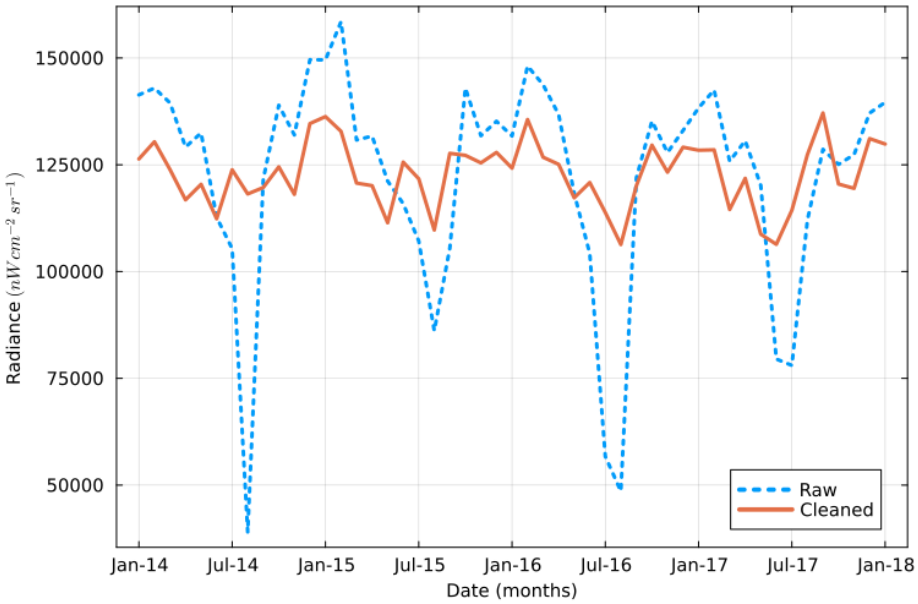


Figure 1: Aggregate radiance of raw and cleaned nighttime lights for Mumbai, January 2014 to January 2018.

Table 1: Effect of cleaning on nighttime lights for Mumbai.

Statistic	Raw	Cleaned
Minimum	-0.14	0.0
Mean	9.1	9.1
Median	3.2	0.0
99th percentile	54.0	59.0
Maximum	4300.0	1700.0

After cleaning there are no negative values, and the maximum is lower because outliers are removed. Half the raw values are below 4, attributable to background noise, and are zeroed. The higher 99th percentile reflects the bias correction.

AI usage disclosure

Generative AI (Anthropic's Claude, through Claude Code) assisted with parts of the test suite, documentation of the outlier filters' limitations, and the continuous integration configuration. The authors reviewed all AI-assisted changes and take responsibility for them. AI assistance was also used in formatting tables in the markdown document.

Acknowledgements

We acknowledge contributions from Anshul Tayal, Siddhant Chauddhary, and Hrishikesh Saikia for their contributions to the codebase and package design, and Kapilan Mahalingam for package improvements, test development, and manuscript preparation. This work received financial support from the MIT Julia Lab.

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